1. **At least 3–5 core figures or tables**

Table 1: Tariff Exposure and Effective Tax Rate

Dependent variable: Winsorized ETR (p1/p99). All specifications include firm and year FE.
Standard errors clustered at NAICS 3-digit level (24 industries). Wild bootstrap p = 0.009.

Specification	Coef.	SE	N
Main (ETR + TCJA control)	**−68.0*** **	(15.0)	4,467
R1: No controls	−66.4***	(11.8)	5,332
R1b: No TCJA control	−68.1***	(15.0)	4,467
R2: SIC1 × year FE	−61.2***	(16.9)	4,461
R3: NAICS2 × year FE	−39.6***	(13.8)	4,466
R4: NAICS2 linear trends	−51.7	(32.1)	4,467
R5: Placebo (2017)	+85.2	(50.8)	1,418
R6: Balanced panel	−66.2***	(15.2)	3,826
R7: ETR p5/p95	−24.3**	(9.7)	4,467
R8: ETR [0, 100]	−19.4	(12.0)	4,354
R9: ETR [0, 60]	−8.3	(9.0)	4,252
R11: COVID interaction	−78.6***	(14.0)	4,467
tariff × covid	+30.2**	(14.5)	
R12: Excl. COVID years	−84.9***	(16.9)	3,445

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

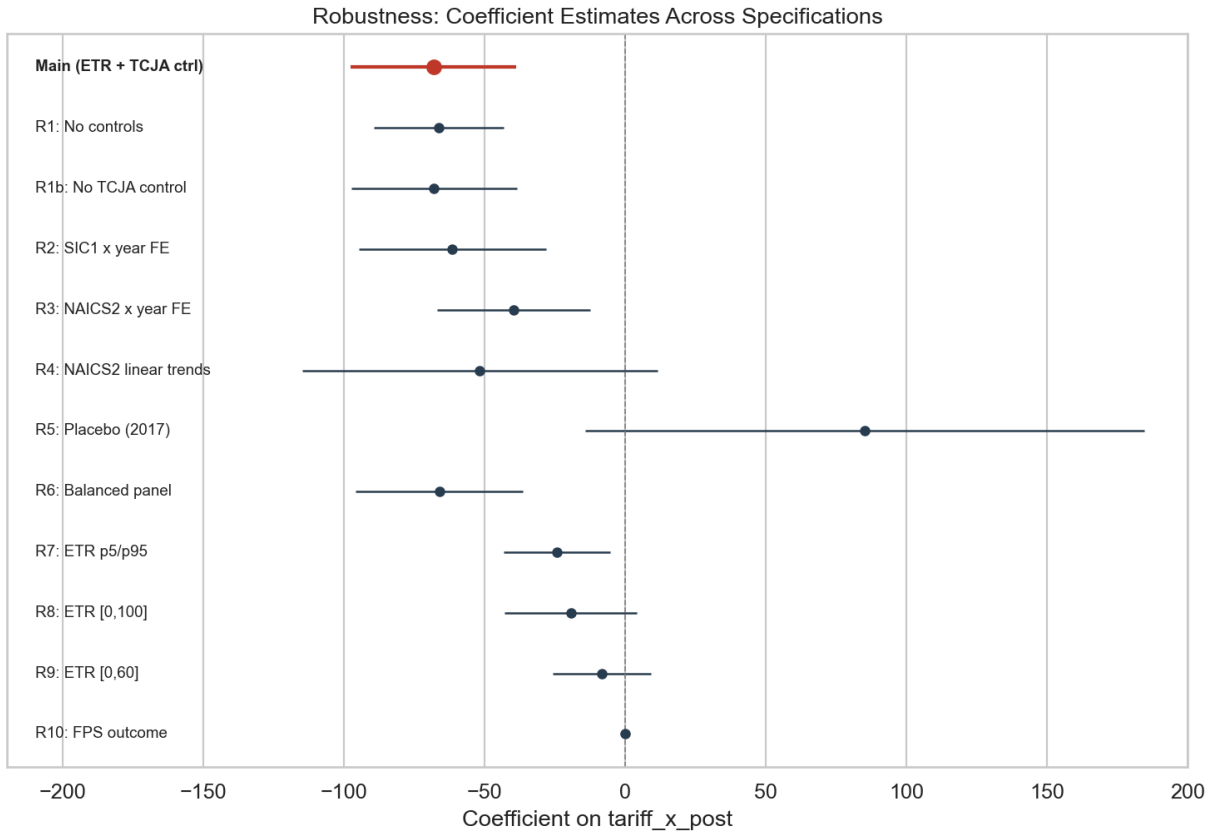
Notes: Main specification includes controls for log revenue, R&D intensity, leverage, and TCJA exposure (pre-treatment foreign profit share × post-TCJA). Placebo uses 2015–2018 data with fake treatment at 2017.

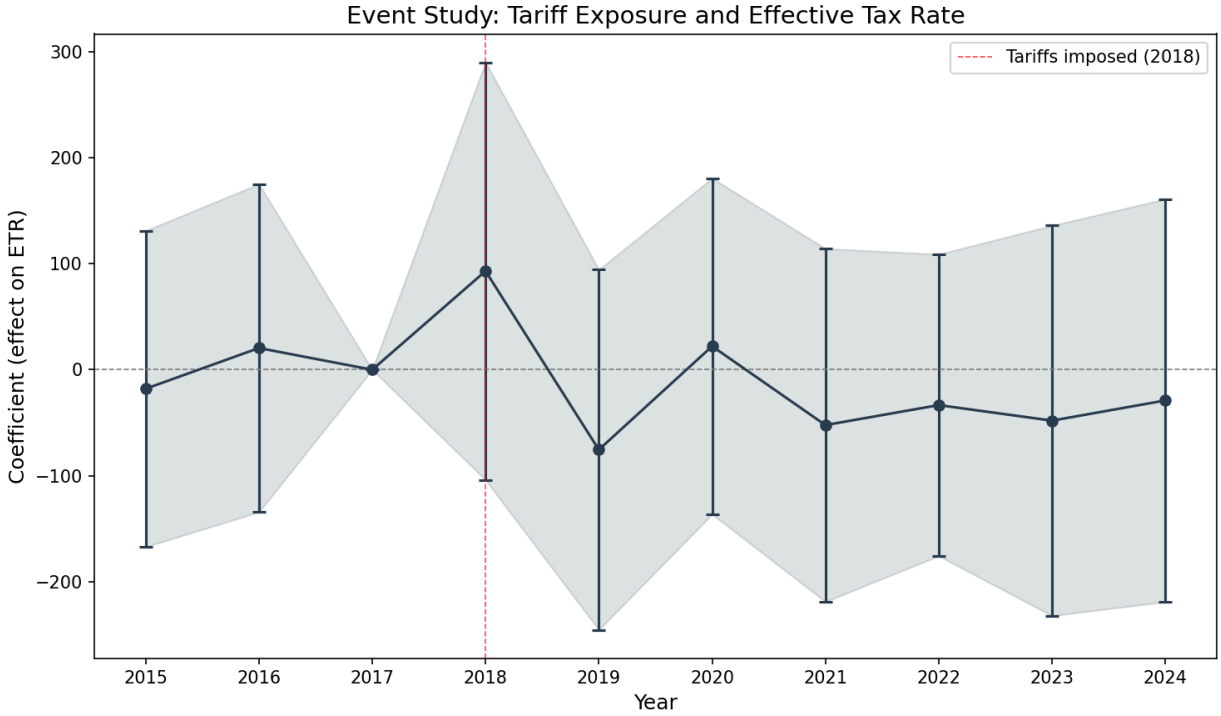
Table 2: Alternative Tariff Exposure Measures

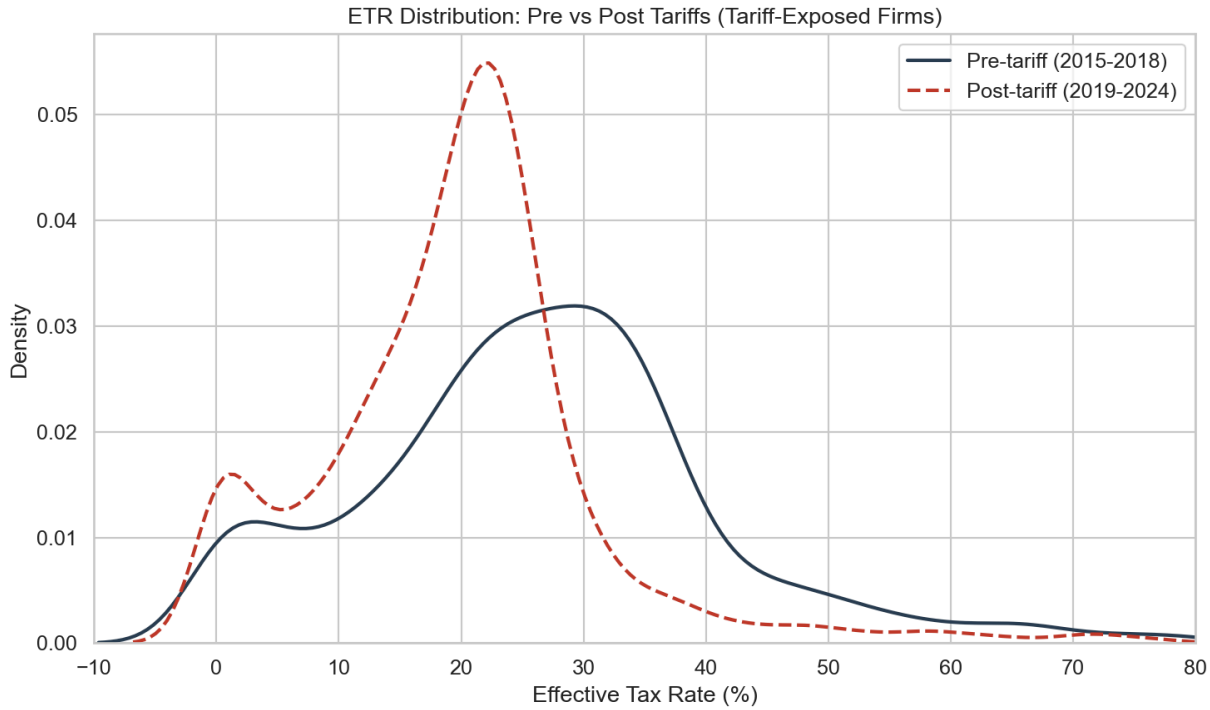
All measures z-scored. Coefficients = effect of 1 SD increase in exposure on winsorized ETR.

Tariff Measure	Coef.	SE	p-value
Mean tariff increase	**−2.9*** **	(0.6)	<0.001
N products targeted	+1.0	(1.2)	0.388
N varieties targeted	+0.4	(0.4)	0.369
SD tariff increase	**−2.0** **	(0.8)	0.024

*** $p < 0.01$, ** $p < 0.05$. Same controls, firm FE, and year FE as main specification. N = 4,467.


Robustness: Coefficient Estimates Across Specifications
Main (ETR + TCJA ctrl)
R1: No controls
R1b: No TCJA control
R2: SIC1 x year FE
R3: NAICS2 x year FE
R4: NAICS2 linear trends
R5: Placebo (2017)
R6: Balanced panel
R7: ETR p5/p95
R8: ETR [0,100]
R9: ETR [0,60]
R10: FPS outcome
−200
−150
−100
−50
0
50
100
150
200
Coefficient on tariff_x_post


Event Study: Tariff Exposure and Effective Tax Rate
Tariffs imposed (2018)
Coefficient (effect on ETR)
300
200
100
0
−100
−200
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
Year



2. **All code producing those results**

https://github.com/amritkgill/tariffs-profit-shifting

3. **A short interpretation (1–2 pages) explaining patterns, sign, magnitude, and** plausibility

The main regression coefficient is -68.0, which means for every 1 unit increase in mean tariff exposure, effective tax rate drops by 68 percentage points after 2018. But no one has a mean tariff increase of 100%. The regression doesn't know specific industries, and it just sees the tariff rate and tells the relationship between that number and ETR. -68 is just the main basic regression for the overall effect, while multiplying it to specific industry tariff rates actually yields the accurate effect of tariffs on ETR.

The mean tariff increase ranges from 4% to 21%. The most exposed tariff industry was Computer and Electronics at 21%, so the actual effect would be the effective tax rate dropping by 14.3 percentage points (0.21 × -68.0 = -14.3). For a less exposed industry like Furniture, the effective tax rate would drop by 2.7 percentage points (0.04 × -68.0 = -2.7). The p value is less than 0.001, which means there is less than 0.1% chance this happened randomly. The main caps extreme values at the 1st and 99th percentiles, and cuts off the most extreme 2% of values. The conservative estimate caps the values at the 5th and 95th percentiles, and cuts off the most extreme 10%. When doing the conservative estimate,

the coefficient shrinks from -68 to -24.3. The real world effect for Computers and Electronics goes from -14.3pp to -5.1pp, and for Furniture it goes from -2.7pp to -0.972.

The ETR data has firms paying anywhere from -10% to 80%. Some firms have really extreme tax rates, like negative ETRs where they got more back in tax credits than they paid, or very high ETRs (above 60%). Some robustness specs that trim ETR to narrower ranges (0-100 and 0-60) lose significance. This is because trimming removes the firms with the biggest ETR swings, which are the ones doing the most profit shifting.

Both the main and the conservative regression are economically meaningful and consistent with profit shifting. If a firm's ETR drops 5 pp, that means they paid 50 million less in taxes, which is significant. The main estimate says it could be as large as 140 million per billion of income. Tariffs raise costs for exposed firms and they look for ways to offset the costs. One way is to shift profits to low-tax countries, and when profit is moved abroad, less income is taxed at U.S. rates, which is what leads to the ETR drop. If firms weren't profit shifting, then they'd just absorb tariff costs, and there'd be no change in ETR. The fact that ETR drops is evidence that they're profit shifting.

For plausibility, the event study is consistent with the parallel trends assumption. The pre-period coefficients for 2015 and 2016 are not statistically distinguishable from zero, but the estimates are imprecise because there are only 24 industry clusters. The placebo test is stronger evidence. When I pretended tariffs started in 2017 using only 2015-2018 data, the coefficient was +85.2 and not significant ($p = 0.107$), which confirms there was no pre-existing trend. If something else was driving the result, the placebo would also be negative, but it's not.

The timing also lines up. The effect starts in 2019, which is the first full fiscal year under tariffs. It doesn't show up in 2017 or earlier, which rules out the possibility that some pre-existing trend is behind the result.

I tested 4 different ways to measure how hard tariffs hit an industry. I looked at the mean tariff rate, number of products targeted, number of varieties targeted, and how spread out the tariff rates were within the industry. Only the mean tariff rate and the spread of tariff rates mattered for predicting the effective tax rate. The mean tariff increase is highly significant ($p < 0.001$), but the number of products and varieties targeted are both insignificant. This makes sense because firms respond to how expensive the tariff is, not how many product lines got hit. The incentive to shift profits depends on the size of the cost shock, not the breadth.

I also checked whether the 2017 Tax Cuts and Jobs Act is confounding my result, since TCJA changed international tax rules at nearly the same time as tariffs. The correlation between tariff exposure and pre-treatment foreign profit share is only 0.03, so high-tariff industries don't systematically have more foreign operations. The TCJA control coefficient

is basically zero and not significant, and adding or removing it barely changes the main result (-68.0 vs -68.1).

I found that COVID was actually weakening my result. When I added a tariff × COVID interaction for 2020-2021, the main coefficient went from -68.0 to -78.6, and the COVID interaction was +30.2 ($p = 0.048$). COVID partially reversed the profit-shifting effect during the pandemic, probably because factories shut down, firms lost money, and got government relief. When I excluded 2020-2021 entirely, the coefficient was -84.9. So my baseline of -68.0 is actually the conservative estimate.

4. **A robustness strategy outline for next steps**

I've already run several robustness checks, and my main result holds across most of them. The coefficient stays negative and significant when I remove controls, add industry-by-year fixed effects, use a balanced panel, and control for COVID. The placebo test passes, and the event study shows clean pre-trends.

I plan on running a leave-one-industry-out test, where I drop one NAICS-3 industry at a time and re-run the main regression. My most exposed industry is Computers and Electronics, which makes up a large share of the sample. I want to make sure that dropping it doesn't kill the result. If the effect survives without it, that means multiple industries are contributing to the finding, not just that one.

I also want to explore heterogeneity. Currently, my regression says tariff-exposed firms lowered their ETRs , but it treats all tariff-exposed firms the same. A giant company like Apple and a small manufacturer both get the same tariff score if they're in the same industry. I want to know which firms within the tariff-exposed group shifted the most. Specifically, I want to interact tariff exposure with firm size and R&D intensity to see if larger firms or more R&D-intensive firms shift more aggressively. The preliminary tests show that R&D intensity × post is significant, which means that firms with more R&D did see bigger ETR drops after 2018. This supports profit shifting because R&D-heavy firms have the resources to shift profits, like patents or intellectual property, so it makes sense they'd respond more aggressively to the tariff cost shock. If the effect were the same for all firms regardless of size or R&D, it would be harder to argue it's profit shifting specifically. But the fact that it's concentrated in firms with more tax planning capability makes the profit-shifting explanation more convincing.

One limitation is that I don't have firm-level import data. Every firm in the same industry gets the same tariff score, even though some firms import way more from China than others. This makes my estimate weaker because the regression can't tell the heavily-hit firms apart from the barely-hit ones, so the coefficient gets pulled toward zero. The true effect is probably larger than -68. If I had data on how much each firm specifically imports

from China, the estimate would be more precise, but I'll address this limitation in the paper.